THESIS REVIEWER'S REPORT

I. IDENTIFICATION DATA

Thesis title:	**Exploring Regret Minimization and Its Variants in Normal-Form Games**
Author's name:	**Rozálie Bílková**
Type of thesis :	bachelor
Faculty/Institute:	Faculty of Electrical Engineering (FEE)
Department:	Department of Cybernetics
Thesis reviewer:	Lukáš Adam
Reviewer's department:	University of West Bohemia in Pilsen

II. EVALUATION OF INDIVIDUAL CRITERIA

Assignment — **challenging**

How demanding was the assigned project?

I consider this bachelor thesis as challenging. The difficulty comes from the sheer overview of the many concepts. It requires both a deep mathematical understanding and good programming skills.

Fulfilment of assignment — **fulfilled**

How well does the thesis fulfil the assigned task? Have the primary goals been achieved? Which assigned tasks have been incompletely covered, and which parts of the thesis are overextended? Justify your answer.

All bullets in the assignment were fulfilled.

Methodology — **correct**

Comment on the correctness of the approach and/or the solution methods.

The selected approach was relatively straightforward yet correct. First, define the problem. Second, define the solution concepts. Third, define the optimization methods. Fourth, combine the previous parts in a numerical study.

Technical level — **C - good.**

Is the thesis technically sound? How well did the student employ expertise in the field of his/her field of study? Does the student explain clearly what he/she has done?

All individual chapters are written properly. The definitions seem to be correct. However, the weakest point is in the combination of chapters, which is practically non-existent. The notation differs across chapters (for example using l and g for the loss function), the topic changes (in Chapter 2, the thesis first analysis general games, then two-person zero-sum games and then general games again), some topics are not defined (the crucial topic of exploitability in Chapter 4 is not defined in the previous chapters), the flow is sometimes not logically consistent (re-definition of matrix A in Section 2.5), the chapters are not connected (there are little or no references across chapters), ...

The provided Julia code is of a mixed quality. There was lots of effort into writing it. On the other hand, the repository is missing commits, there are no tests, there is a file named ugh.jl submitted by mistake and even a required file (exploitability.jl) is missing. Some parts (such as Game) are not the way Julia code should be written (too many elseifs).

Formal and language level, scope of thesis — **A - excellent.**

Are formalisms and notations used properly? Is the thesis organized in a logical way? Is the thesis sufficiently extensive? Is the thesis well-presented? Is the language clear and understandable? Is the English satisfactory?

I appreciate that the thesis was written in English. There are some typos (unfortunately, the first one appears at line 4 of the introduction), but, otherwise, the thesis is simple to read and understand. The glossary of terms at the beginning is very helpful.

Selection of sources, citation correctness	A - excellent.
Does the thesis make adequate reference to earlier work on the topic? Was the selection of sources adequate? Is the student's original work clearly distinguished from earlier work in the field? Do the bibliographic citations meet the standards?	
The literature is correctly selected and cited.	

Additional commentary and evaluation (optional)
Comment on the overall quality of the thesis, its novelty and its impact on the field, its strengths and weaknesses, the utility of the solution that is presented, the theoretical/formal level, the student's skillfulness, etc.
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III. OVERALL EVALUATION, QUESTIONS FOR THE PRESENTATION AND DEFENSE OF THE THESIS, SUGGESTED GRADE

Summarize your opinion on the thesis and explain your final grading. Pose questions that should be answered during the presentation and defense of the student's work.

The grade that I award for the thesis is **B - very good.**

In my opinion, the biggest drawbacks of the thesis come from the fact that the candidate attempted to present too many concepts (the submitted thesis has 64 pages) and there was not enough space to go into details for any of them. Instead of this approach, I would very much prefer to focus on a few selected topics and write thesis with a coherent story, where the chapters are connected. The lack of simple explanatory examples makes it difficult to judge to which extent the candidate understood the topic.

Even though I was relatively critical in some parts of this review, the candidate has definitely shown that she is capable of all the things required for bachelor's degree, including understanding mathematical concepts, reading scientific papers, replicating results and writing them down. For this reason, I highly recommend the thesis for defense.

I have the following questions:

- The thesis states that when a player plays any strategy (possibly pure), he obtains the loss for *all* his pure strategies. This seems very restrictive. Is it really needed?
- What is an example of a game which falls into the considered framework, that is two-person zero-sum game, which uses mixes strategies and is played repeatedly?
- Give some examples when the subset chain of PNE, MNE, CE and CCE is satisfied as strict inequalities.
- What is the connection between the mixed Nash equilibrium and Theorem 1?

Date: **18.5.2026** Signature: